

```

import pandas as pd

df = pd.read_csv("/content/Mall_Customers.csv")
df.head()

{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 200,\n  \"fields\": [\n    {\n      \"column\": \"CustomerID\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 57,\n        \"min\": 1,\n        \"max\": 200,\n        \"num_unique_values\": 200,\n        \"samples\": [\n          96,\n          16,\n          31\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Gender\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"Female\",\n          \"Male\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Age\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 13,\n        \"min\": 18,\n        \"max\": 70,\n        \"num_unique_values\": 51,\n        \"samples\": [\n          55,\n          26\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Annual Income (k$)\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 26,\n        \"min\": 15,\n        \"max\": 137,\n        \"num_unique_values\": 64,\n        \"samples\": [\n          87,\n          101\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Spending Score (1-100)\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 25,\n        \"min\": 1,\n        \"max\": 99,\n        \"num_unique_values\": 84,\n        \"samples\": [\n          39,\n          83\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    ]\n  }},\n  \"type\": \"dataframe\",\n  \"variable_name\": \"df\"}

X = df[["Annual Income (k$)", "Spending Score (1-100)"]]

from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

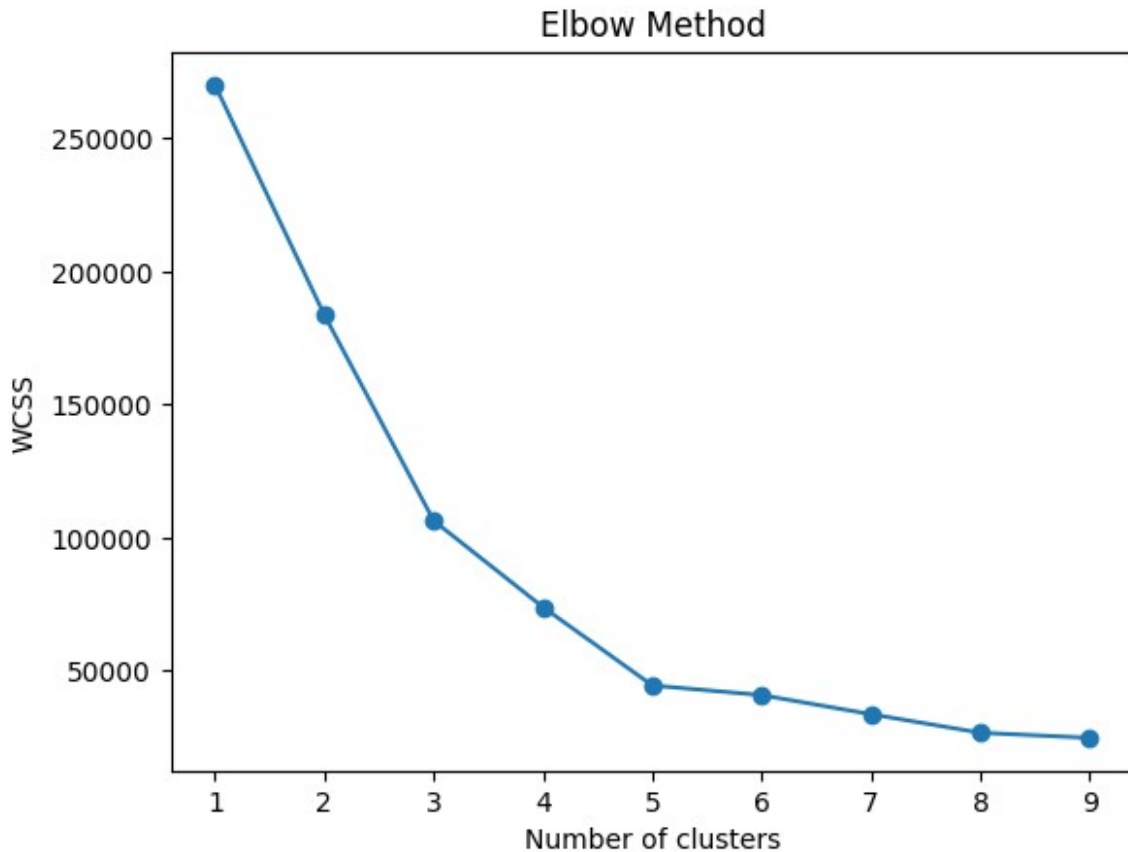
wcss = []

for i in range(1, 10):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 10), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of clusters")

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plt.ylabel("WCSS")
plt.show()
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```
kmeans = KMeans(n_clusters=3, random_state=42)
df["Cluster"] = kmeans.fit_predict(X)

df.head()

{"summary":{"\n  \"name\": \"df\", \n  \"rows\": 200, \n  \"fields\": [\n    {\n      \"column\": \"CustomerID\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 57, \n        \"min\": 1, \n        \"max\": 200, \n        \"num_unique_values\": 200, \n        \"samples\": [\n          96, \n          16, \n          31\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }, \n      \"column\": \"Gender\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 2, \n        \"samples\": [\n          \"Female\", \n          \"Male\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }, \n      \"column\": \"Age\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 13, \n        \"min\": 18, \n        \"max\": 70, \n        \"num_unique_values\": 51, \n        \"samples\":
```

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[\\n          55,\\n          26\\n          ],\\n          \\\"semantic_type\\\":
\\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\n          },\\n          {\\n
\\\"column\\\": \\\"Annual Income (k$)\\\",\\n          \\\"properties\\\": {\\n
\\\"dtype\\\": \\\"number\\\",\\n          \\\"std\\\": 26,\\n          \\\"min\\\": 15,\\n
\\\"max\\\": 137,\\n          \\\"num_unique_values\\\": 64,\\n
\\\"samples\\\": [\\n          87,\\n          101\\n          ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\n
          },\\n          {\\n          \\\"column\\\": \\\"Spending Score (1-100)\\\",\\n
\\\"properties\\\": {\\n          \\\"dtype\\\": \\\"number\\\",\\n          \\\"std\\\":
25,\\n          \\\"min\\\": 1,\\n          \\\"max\\\": 99,\\n
\\\"num_unique_values\\\": 84,\\n          \\\"samples\\\": [\\n          83,\\n
39\\n          ],\\n          \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n          }\\n          },\\n          {\\n          \\\"column\\\":
\\\"Cluster\\\",\\n          \\\"properties\\\": {\\n          \\\"dtype\\\": \\\"int32\\\",\\n
\\\"num_unique_values\\\": 3,\\n          \\\"samples\\\": [\\n          2,\\n
1\\n          ],\\n          \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n          }\\n          }\\n          ]\\n
n}\\\", \"type\": \"dataframe\", \"variable_name\": \"df\"}

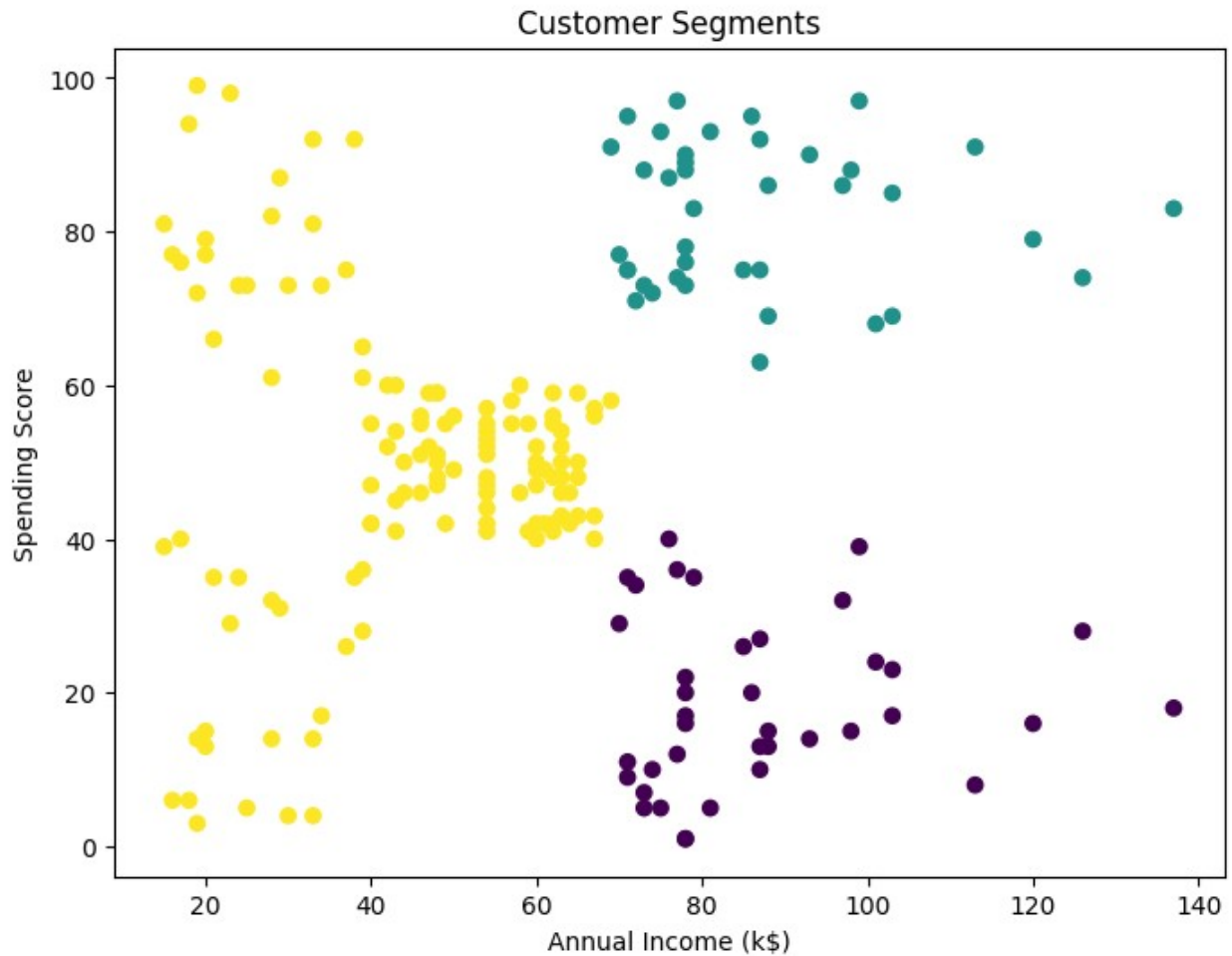
plt.figure(figsize=(8,6))

plt.scatter(X.iloc[:,0], X.iloc[:,1], c=df[\"Cluster\"]

plt.xlabel(\"Annual Income (k$)\")
plt.ylabel(\"Spending Score\")
plt.title(\"Customer Segments\")

plt.show()

```



```
print("Cluster Centers:")
print(kmeans.cluster_centers_)
```

```
Cluster Centers:
[[87.          18.63157895]
 [86.53846154 82.12820513]
 [44.15447154 49.82926829]]
```

```
print(df.info())
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 6 columns):
#   Column              Non-Null Count  Dtype
---  -
0   CustomerID          200 non-null   int64
1   Gender              200 non-null   object
2   Age                 200 non-null   int64
3   Annual Income (k$)  200 non-null   int64
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4    Spending Score (1-100)    200 non-null    int64
5    Cluster                    200 non-null    int32
dtypes: int32(1), int64(4), object(1)
memory usage: 8.7+ KB
None

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      CustomerID      Age  Annual Income (k$)  Spending Score (1-
100) \
count    200.000000    200.000000          200.000000
200.000000
mean     100.500000    38.850000          60.560000
50.200000
std       57.879185    13.969007          26.264721
25.823522
min        1.000000    18.000000          15.000000
1.000000
25%       50.750000    28.750000          41.500000
34.750000
50%      100.500000    36.000000          61.500000
50.000000
75%      150.250000    49.000000          78.000000
73.000000
max      200.000000    70.000000         137.000000
99.000000

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      Cluster
count    200.000000
mean      1.425000
std       0.792157
min        0.000000
25%        1.000000
50%        2.000000
75%        2.000000
max        2.000000

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```
df["Gender"] = df["Gender"].map({"Male": 0, "Female": 1})
```

```
from sklearn.preprocessing import StandardScaler
```

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scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
print(df["Cluster"].value_counts())
```

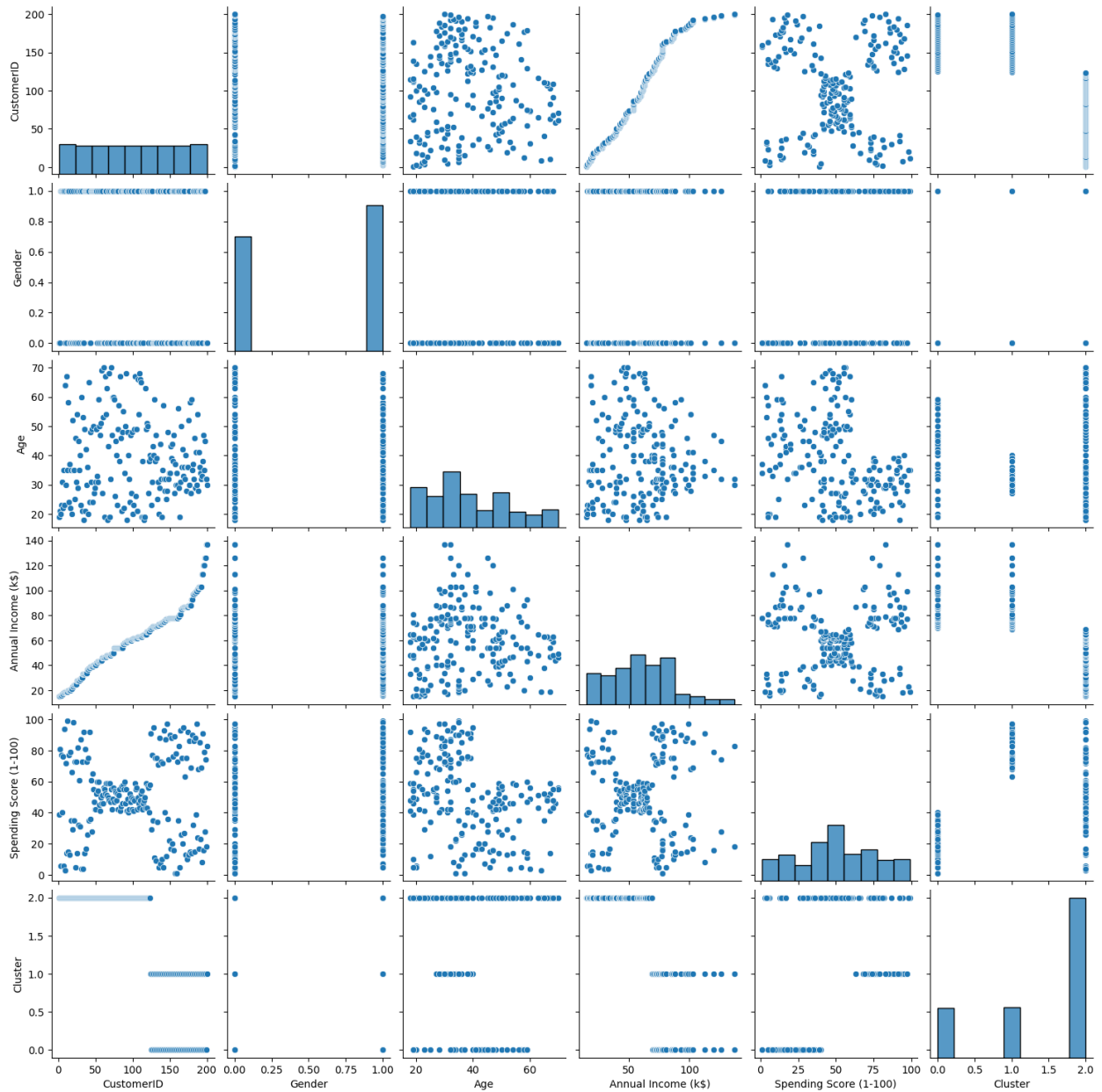
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Cluster
2     123
1      39
0      38
Name: count, dtype: int64

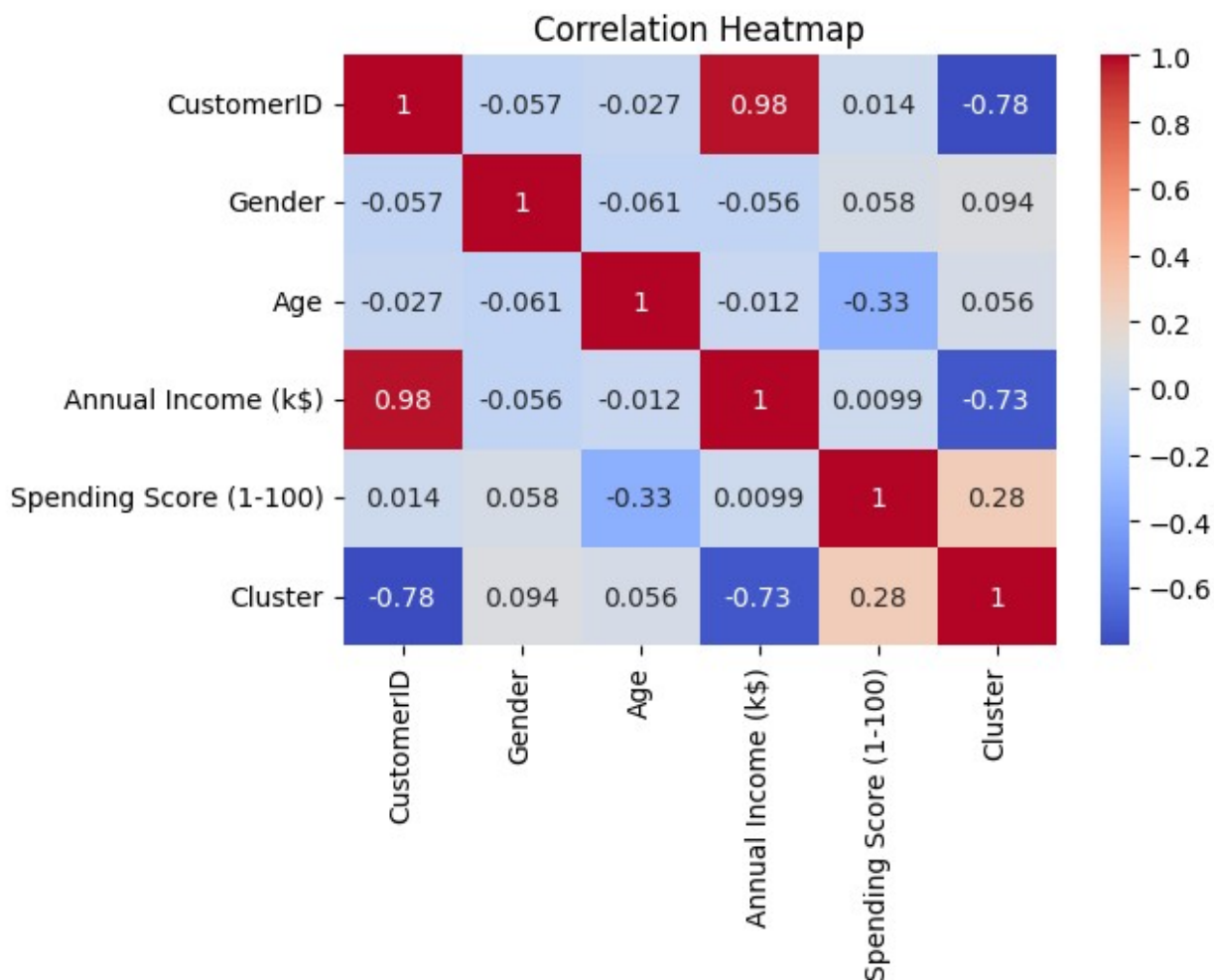
```

```
df.to_csv("segmented_customers.csv", index=False)
```

```
import seaborn as sns
sns.pairplot(df)
plt.show()
```



```
plt.figure(figsize=(6,4))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



```
from sklearn.metrics import silhouette_score

score = silhouette_score(X_scaled, df["Cluster"])
print("Silhouette Score:", score)

Silhouette Score: 0.46658474419000145

from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.scatter(df["Age"], df["Annual Income (k$)"], df["Spending Score (1-100)"],
          c=df["Cluster"])

ax.set_xlabel("Age")
ax.set_ylabel("Income")
ax.set_zlabel("Spending Score")
```

```
plt.show()
```

